

Rewards:

$$A=3$$

$$B=6$$

$$C=4$$

Since each machine is played once:

$$N_A = N_B = N_C = 1$$

Initial Q -values:

$$Q_A = 0 + \frac{3-0}{1} = 3$$

$$Q_B = 0 + \frac{6-0}{1} = 6$$

$$Q_C = 0 + \frac{4-0}{1} = 4$$

Therefore:

$$Q = [3, 6, 4]$$

Highest Q -value = B.

Play 1 - B gets 7

B was already played once so:

$$N_B = 2$$

$$Q_B = 6 + \frac{7-6}{2}$$

$$= 6 + 0.5$$

$$Q_B = 6.5$$

$$\text{Now: } Q = [3, 6.5, 4]$$

Play 2 - B gets 5

B has now been played 3 times:

$$N_B = 3$$

$$Q_B = 6.5 + \frac{5-6.5}{3}$$

$$Q_B = 6.5 - \frac{1.5}{3} = 6$$

$$\text{Final: } Q = [3, 6, 4]$$

Final matrices:

$$N = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R = \begin{bmatrix} 3 & 0 & 0 \\ 6 & 7 & 5 \\ 4 & 0 & 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} 3 & 3 & 3 \\ 6 & 6.5 & 6 \\ 4 & 4 & 4 \end{bmatrix}$$

$$\text{Total reward: } 3 + 6 + 4 + 7 + 5 = 25$$

$$\text{Average reward: } \frac{25}{5} = 5.0$$